

COURSE 3 - DATA ENGINEERING 2026

C3-09 - Production Capstone & Engineering Transition

Final core phase of the Four-Course Course 3 curriculum. Integrate prior skills; do not replace earlier environments or skip evidence.

Artifact	C3-09 START HERE
Status	2026 - LEARNER EDITION
Phase	C3-09 Production Capstone & Engineering Transition 15 lessons 36-48 focused hours
Prerequisite	C3-08 Gate PASS; all earlier C3 evidence retained.
Assessment boundary	C3-104 is the phase Gate only. C3-113 remains the later course-wide Final Production Gate.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: requirement -> mental picture -> small proof -> guided integrated build -> expected result -> why it worked -> break/fix -> changed independent case -> validation -> production reasoning -> evidence. Integration is not mastery until the learner can explain and recover the system independently.

Exact lesson sequence

C3-09-L01 - Capstone ADR: business problem, architecture and non-functional requirements

C3-09-L02 - Source contracts, synthetic/public data and reproducible inputs

C3-09-L03 - Build the batch ingestion path

C3-09-L04 - Add incremental / CDC-style change handling

C3-09-L05 - Design layered lakehouse and serving architecture

C3-09-L06 - Transformation engineering, modeling and tests

C3-09-L07 - Orchestration, retries, backfills and partial reruns

C3-09-L08 - Integrate a real Spark component

C3-09-L09 - Quality gates and source-to-output reconciliation

C3-09-L10 - Observability, SLOs and operational monitoring

C3-09-L11 - Security, governance and privacy-safe evidence

C3-09-L12 - CI/CD, deployment and controlled change

C3-09-L13 - Incident injection, recovery and prevention

C3-09-L14 - Runbook, README, handoff and portfolio evidence

C3-09-L15 - Final architecture defense and engineering transition

Learner route

C3-100 lesson -> matching C3-101 RetailPulse checkpoint -> save evidence -> matching C3-102 protected review only if needed -> changed retry -> next lesson. After L15, complete C3-103 TransitOps. Then attempt C3-104A EnergyOps independently.

Environment rule

Reuse C3-02 dbt/DuckDB Python 3.13 environment when dbt is exercised; reuse the prior Airflow environment for orchestration; reuse C3-05 Spark environment for Spark 4.2; reuse prior Kafka/Delta/Fabric environments if you elect the live integration path. C3-09 itself adds no new mandatory installation.

Current 11 September 2026 integration snapshot

Python 3.14.7 is the current 3.14 maintenance release. dbt-core 1.12.4 is the current stable Core release (released 8 September 2026). C3-02 intentionally keeps its dbt-duckdb 1.11 compatibility environment for that local lab. Airflow 3.3.1, Spark 4.2.0, Delta Lake 4.4.0 and Kafka 4.3.1 remain the current course lanes. GitHub environments/OIDC and OneLake control-plane/data-plane security remain current deployment/security authorities.

Critical boundary

Passing C3-104 completes the C3-09 phase only. It does NOT graduate Course 3. C3-113 is the separate later Final Production Gate and remains protected until whole-course reconciliation/Mentor routing is ready.